

From Solomonoff to Ontology: Algorithmic Probability as Cosmological Measure

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Abstract

Solomonoff's algorithmic probability provides the unique optimal prior for inductive inference. I argue it also governs the structure of reality itself. The extension requires two steps: first, establish that reality *is* mathematical structure via an exhaustive pattern-randomness dichotomy; second, show that algorithmic probability is the natural measure over structure-space. The result transforms Solomonoff's epistemic insight into ontological fact. Key implications include: (1) non-computable structures have measure zero, deriving rather than assuming the restriction to computable universes; (2) the multiverse measure problem is solved by weighting structures via $P(S)$ proportional to 2 raised to negative $K(S)$; (3) Occam's Razor is grounded in ontology. The framework explains why simpler theories are more likely true: they describe higher-measure structures. We reason correctly about reality because algorithmic probability governs both inference and existence.

1. Introduction

Solomonoff (1964) established that algorithmic probability provides the unique optimal framework for inductive inference. Given data D , the probability of hypothesis H should be weighted by 2 raised to negative $K(H)$, where $K(H)$ is the Kolmogorov complexity of H . This prior dominates all computable alternatives in the sense that no computable predictor can systematically outperform it.

This paper argues for a stronger claim: algorithmic probability governs the structure of reality itself, as well as reasoning about it. The measure over mathematical structures is $P(S)$ proportional to 2 raised to negative $K(S)$. Solomonoff's epistemic principle reflects ontological fact.

The argument proceeds in two stages. Section 2 establishes that reality is mathematical structure via the pattern-randomness dichotomy. Section 3 shows that algorithmic probability is the unique natural measure over structure-space. Sections 4 and 5 develop implications for physics and philosophy.

2. Reality as Mathematical Structure

2.1 The Dichotomy

Any intelligible reality exhibits either discoverable patterns or apparent randomness. This dichotomy is exhaustive.

Patterns are regularities expressible through mathematical relations. Randomness is characterized via probability distributions and algorithmic incompressibility. A string s is random if and only if $K(s)$ is approximately equal to the length of s : no compression is possible, no shorter description exists.

Both categories are inherently mathematical. There is no third option. For something to exist determinately, it must have features distinguishing it from nonexistence. Those features either exhibit regularity (pattern) or resist compression (randomness). Either way, they fall within mathematical description.

2.2 From Isomorphism to Identity

Physical structures are isomorphic to mathematical structures by definition of “physical structure.” What distinguishes them? Not their relationships (identical by isomorphism). Not causal powers (supervening on relations). Not observables (reducing to relations).

The only candidate is non-relational substrate. But substrate without properties is indistinguishable from nothing. By the Identity of Indiscernibles, physical and mathematical structure are identical. Reality is mathematical structure, identical to it.

3. The Simplicity Dominance Principle

The philosophical derivation of SDP from mathematical structural realism is given in Bernstein (2026d).

3.1 The Measure Problem

If reality is mathematical structure, all self-consistent structures exist. This raises the measure problem: among all structures, which should observers expect to inhabit? Note that inconsistent descriptions exist as information-bearing patterns: the string “ $2+2=5$ ” has a well-defined Kolmogorov complexity and occupies a definite location in string-space. Every medium encoding it is a consistent mathematical structure, which is why the description exists. But the arithmetic claim it encodes has no consistent model under the standard interpretation. The description is compressible information; the described state of affairs is not instantiable. Repairing the claim requires redefining terms through consistent reasoning. Inconsistent content can only be made coherent by consistent mathematics: reason is the sole construction medium.

Since reality is structure, the measure is discovered. We seek the natural measure over structure-space.

3.2 Formulation

I propose:

SDP: $P(S)$ proportional to 2 raised to negative $K(S)$

where $K(S)$ is the Kolmogorov complexity of S 's generating rules.

3.3 Why Solomonoff's Prior?

Algorithmic probability is uniquely motivated:

Dominance: Solomonoff's completeness theorem establishes that algorithmic probability dominates all computable measures. No computable alternative systematically outperforms it.

Invariance: The measure is machine-independent up to multiplicative constants. Different universal Turing machines yield K values differing by at most an additive constant, absorbed by the exponential.

Combinatorics: In a prefix-free coding, each program of length k occupies a fraction 2^{-k} of program-space. A 10-bit program occupies fraction 2^{-10} . A 100-bit program occupies fraction 2^{-100} . Shorter programs have exponentially higher measure, not because they are more numerous but because each claims a larger fraction of program-space. A "program" here is a finite description, not a process. No computation is required. The structures exist as mathematics. The measure is over specifications, not executions.

Coherence: This explains why Occam's Razor works. If simpler theories were not more likely true, parsimony's success in the history of science would be miraculous.

The critic asking "why this measure?" must produce an alternative satisfying comparable desiderata. Lebesgue measure is undefined over discrete program space. Uniform measure requires arbitrary cutoffs. Complexity-favoring measures yield the Boltzmann brain catastrophe.

Algorithmic probability is not one choice among many. It is the unique natural measure satisfying information-theoretic requirements.

4. Implications for Physics

4.1 Non-Computable Structures Have Measure Zero

Non-computable structures require oracle machines encoding infinite brute facts. If oracles had patterns, they would be computable. So $K(\text{oracle})$ equals infinity,

yielding P equal to 2 raised to negative infinity, which equals zero.

This derives Schmidhuber's restriction to computable universes rather than assuming it. Non-computable structures exist mathematically but contribute nothing to predictions.

4.2 Structures with Simple Laws Dominate

Algorithmic probability weights structures by the length of their shortest generating program. Structures with simple, finitely specifiable laws have low K and high measure. Structures requiring complex or infinite specification have high K and low measure.

This explains the observed simplicity of physical laws. The Standard Model Lagrangian (kinetic minus potential energy for all particles and forces) fits on a t-shirt. General relativity derives from a one-line action principle. If all structures existed with equal measure, such simplicity would be miraculous. Under SDP, simplicity is expected.

4.3 Variational Laws Explained

Physical laws take variational form: least action, Einstein-Hilbert action, path integrals. This explains why.

Consider two ways to specify identical physics: (a) "Systems follow paths extremizing S equal to the integral of $L dt$ " (compact variational principle); (b) exhaustive enumeration of permitted trajectories.

Specification (a) has vastly lower $K(S)$. Variational universes dominate the measure. The principle selects for universes whose physics takes variational form, because variational specifications occupy more of program-space.

4.4 Fine-Tuning Resolved

Physical constants appear calibrated for complexity. This explains: among observer-supporting structures, those with simple laws dominate. Fine-tuning is what simplicity looks like from the inside. No designer is needed because no selection occurred. The appearance of tuning is a measure effect: the simplest observer-permitting structures dominate.

4.5 Unreasonable Effectiveness Explained

Wigner (1960) called the effectiveness of mathematics in physics "unreasonable." Under SDP, it is predicted. Our universe has low K : its laws are simple enough to be discovered by finite minds doing mathematics. Mathematical structures discovered through pure reasoning describe physical reality because our reality has a short generating description. The effectiveness is not unreasonable. It is expected.

4.6 Competing Theories Ranked

SDP applies to the competition between physical theories. Wolfram’s Physics Project proposes that the universe emerges from extremely simple hypergraph rewriting rules, potentially expressible in bits rather than kilobits. This aligns with SDP’s prediction. String theory provides a contrasting test: its total specification complexity (approximately 3,000 to 4,700 bits) dwarfs any plausible generating rule for the Standard Model’s 19 to 26 scalar parameters, rendering it exponentially disfavored (Bernstein, 2026g).

5. From Epistemology to Ontology

Solomonoff intended algorithmic probability as an epistemic tool: how to reason given uncertainty. SDP extends this to ontology: the measure governs existence as well as reasoning.

This extension is not arbitrary. If reality is mathematical structure, then the measure over structures is not imposed from outside but discovered as an intrinsic property of structure-space. Algorithmic probability emerges from the combinatorics of description length.

The result: we reason correctly about reality because algorithmic probability governs both inference and existence. Occam’s Razor works because simpler structures dominate the measure. Scientists tracking parsimony have been unconsciously tracking the natural measure over structure-space.

This completes the Solomonoff program. Algorithmic probability was always ontological; we are now positioned to see why.

6. Relation to Prior Work

Solomonoff (1964): Algorithmic probability for induction. SDP extends to ontology.

Schmidhuber (1997, 2000): The closest precursor. Schmidhuber proposed that the universe is the output of a short program, weighted by algorithmic probability, and explored “algorithmic theories of everything” and the “Speed Prior” (favoring structures computable quickly). Three differences are substantive:

First, Schmidhuber assumes computability as a constraint on which structures exist. SDP derives it: non-computable structures have undefined algorithmic probability and therefore measure zero. The restriction to computable universes is a consequence, not a premise. This matters because assuming computability invites the objection “why only computable structures?” while deriving it closes the question.

Second, Schmidhuber’s framework is primarily epistemic and computational. He asks what a rational agent should predict given data. SDP makes the ontological claim: algorithmic probability governs existence, not just inference. We reason correctly about reality because the same measure governs both. This is a stronger claim that requires the PRD for justification; without establishing that reality IS mathematical structure, the extension from epistemology to ontology is unmotivated.

Third, Schmidhuber’s Speed Prior adds a computational time penalty beyond algorithmic probability. SDP does not. Whether the Speed Prior is correct is an empirical question about whether our universe’s generating rule is not only short but also fast. SDP is agnostic on this: it predicts simplicity of description, not speed of computation. The Speed Prior is a possible refinement of SDP, not a competitor.

Tegmark (2008): Mathematical Universe Hypothesis. SDP provides the measure Tegmark leaves open.

Hutter (2005): AIXI optimal agent. SDP suggests AIXI’s prior reflects ontological structure, not just epistemic optimality.

6.5 Technical Limitations

Three standard objections to algorithmic probability require explicit acknowledgment.

Uncomputability. Kolmogorov complexity $K(S)$ is uncomputable: no algorithm can determine the shortest program generating an arbitrary string (Li & Vitanyi, 2008). This is true but does not undermine the principle’s utility. We do not need exact values of K . We need the ordering: $K(\text{simple laws}) < K(\text{complex laws}) < K(\text{random configurations})$. This ordering is decidable in practice for the structures relevant to physics. The Standard Model has lower K than a random Lagrangian; general relativity has lower K than an arbitrary metric. Uncomputability limits precision, not applicability.

UTM dependence. $K(S)$ depends on the choice of universal Turing machine, up to an additive constant c (the invariance theorem). For short strings, c dominates and K is meaningless. For the structures relevant to cosmology (physical laws with K in the hundreds or thousands of bits) the additive constant is negligible relative to the complexity differences between competing theories. The framework applies to the domain where it matters.

Normalization. The sum $\sum 2^{-K(S)}$ over all computable structures converges to a value < 1 (Solomonoff’s prior is a semi-measure). This is a feature, not a bug: the “missing” probability mass corresponds to uncomputable structures, which under the principle have measure zero. The semi-measure is the correct

measure over computable mathematical structures, which are the only ones with well-defined K .

These limitations are inherited from algorithmic information theory itself and apply equally to any framework using Solomonoff's prior. They do not uniquely burden the ontological extension proposed here.

7. Conclusion

Solomonoff's algorithmic probability, initially an epistemic principle, reflects the structure of reality. We extended the principle from inference to existence via the pattern-randomness dichotomy establishing that reality is mathematical structure.

The implications are substantial: non-computable structures have measure zero, variational laws dominate, fine-tuning is explained, and Occam's Razor is grounded. We understand why simplicity tracks truth: simpler structures dominate the natural measure over structure-space.

Simplicity is not merely a guide to inference. It is the structure of existence itself.

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